$$M_{21} = \frac{\varepsilon_2}{di_1/dt}$$

$$= \frac{60\times10^{-3}\times0.4}{8}$$

$$M_{21} = 3\times10^{-3} H$$

(ii) Induced emf in the second coil due to the rate of change of current in the first coil is

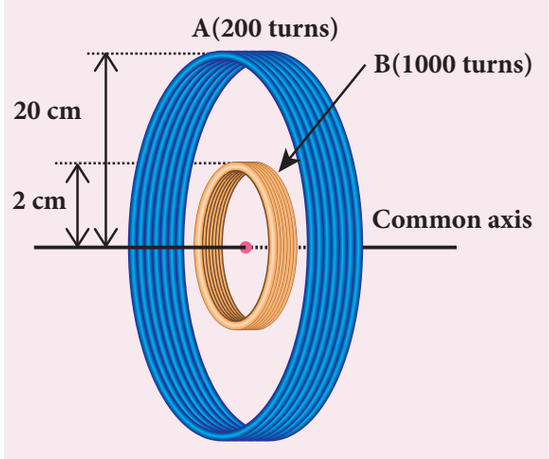
$$\varepsilon_2 = M_{21}\frac{di_1}{dt}$$

$$= \frac{3\times10^{-3}\times12}{0.03}$$

$$\varepsilon_2 = 1.2V$$

EXAMPLE 4.13

Consider two coplanar, co-axial circular coils *A* and *B* as shown in figure. The radius of coil *A* is 20 cm while that of coil *B* is 2 cm. The number of turns is 200 and 1000 for coils *A* and *B* respectively. Calculate the mutual inductance of coil *B* with respect to coil *A*. If the current in coil *A* changes from 2 A to 6 A in 0.04 sec, determine the induced emf in coil *B* and the rate of change of flux through the coil *B* at that instant.



Solution

$N_A = 200$ turns; $N_B = 1000$ turns;
$r_A = 20 \times 10^{-2}$ m; $r_B = 2 \times 10^{-2}$ m;
$dt = 0.04$ s; $di_A = 6-2 = 4$A

Let i_A be the current flowing in coil *A*, then the magnetic field B_A at the centre of the circular coil *A* is

$$B_A = \frac{\mu_\circ N_A i_A}{2r_A} = \frac{4\pi\times10^{-7} N_A i_A}{2r_A}$$

$$= \frac{10^{-7}\times2\times3.14\times200}{20\times10^{-2}}\times i_A$$

$$= 6.28\times10^{-4} i_A \ Wbm^{-2}$$

The magnetic flux linkage of coil *B* is

$$N_B\Phi_B = N_B B_A A_B$$

$$= 1000\times6.28\times10^{-4}\times i_A \times3.14\times\left(2\times10^{-2}\right)^2$$

$$= 7.89\times10^{-4} i_A \ \ Wb\ turns$$

The mutual inductance of the coil *B* with respect to coil *A* is

$$M_{BA} = \frac{N_B\Phi_B}{i_A} = 7.89\times10^{-4} H$$

Induced emf in coil *B* is

$$\varepsilon_B = -M_{BA}\frac{di_A}{dt}$$

Considering only the magnitude,

$$\varepsilon_B = \frac{7.89\times10^{-4}\times(6-2)}{0.04}$$

$$\varepsilon_B = \frac{7.89\times10^{-4}\times(4)}{4\times10^{-2}}$$

$$\varepsilon_B = 7.89\times10^{-2} V$$

$$\varepsilon_B = 78.9 mV$$

The rate of change of magnetic flux of coil *B* is

$$\frac{d(N_B\Phi_B)}{dt} = \varepsilon_B = 78.9\ mWbs^{-1}$$

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